

Dynamic Firm Optimization with Learning-by-Importing Under Tariff Uncertainty

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1 Introduction and Motivation

Trade policy, particularly the use of tariffs, plays a central role in shaping economic outcomes by affecting relative prices, firm decisions, and consumer welfare. Tariffs—taxes levied on imported goods—are a common instrument of trade policy. Governments may impose tariffs as a strategic tool to protect domestic industries, retaliate against perceived unfair trade practices, or shift bargaining power in international negotiations. While these objectives can be politically compelling, the economic consequences are complex and often misunderstood.

For instance, before the 1960s, chicken was considered a delicacy in Europe. However, due to post-World War II intensive chicken farming in the United States, chicken quickly became a staple food in the western world. Thus, what was once considered a luxury suddenly dropped in price and became widely accessible. The introduction of American chicken disrupted European chicken farmers. At its peak, the United States captured over half the market of imported chicken in Europe. As such, some governments, namely France and West Germany, imposed high tariffs on American chicken to protect local chicken farmers. However, the United States retaliated by imposing tariffs on potatoes, light trucks, and other goods^[1]. This is known as the "Chicken War," and some of these tariffs still exist today.

A common misconception about tariffs is that they primarily burden foreign exporters or large multinational corporations. In reality, the cost of tariffs is often passed along domestic supply chains, raising input prices for firms that rely on imported goods and, ultimately, increasing prices for consumers. By increasing the effective cost of imported inputs, tariffs disrupt firm-level production decisions, discourage investment, and reduce productivity. These distortions can ripple through the economy, affecting not just short-term pricing but also long-term capital accumulation and productivity growth.

This paper develops a dynamic computational model to examine how tariffs influence a firm's optimal input choices over time — specifically, imports, investment, and resulting output. Rather than focusing only on static effects, the model captures how firms respond intertemporally to changing tariff environments by solving a dynamic optimization problem using Bellman iteration. Using a Cobb–Douglas production function, capital depreciation, and learning-by-importing, we aim to simulate how trade policy shocks reshape the trajectory of firm growth.

2 Economic Model

Consider a small firm that uses capital K and an imported intermediate input M to produce output Y . The production function is Cobb–Douglas:

$$Y = F(K, M) = K^\alpha M^\gamma,$$

with $0 < \alpha < 1$ and $0 < \gamma < 1$.

The imported input has world price $p = 1$, but is subject to an import tariff τ_t . The tariff $\tau_t \in \{\tau_L, \tau_H\}$ follows a Markov process with transition probabilities

$$P = \begin{pmatrix} P_{LL} & P_{LH} \\ P_{HL} & P_{HH} \end{pmatrix},$$

where $P_{ij} = \Pr(\tau_{t+1} = \tau_j \mid \tau_t = \tau_i)$.

Each period the firm chooses imports $M \geq 0$ and investment $X \geq 0$. Capital evolves according to

$$K' = (1 - \delta) K + X + \psi M,$$

with depreciation rate $\delta \in [0, 1]$. Since a unit of imports embodies foreign know-how or specialized equipment, a share ψ is useful for next-period's capital, augmenting the capital base. The per-period profit is then

$$\Pi(K, M, X; \tau) = K^\alpha [(1 - \psi)M]^\gamma - (1 + \tau)pM - X.$$

Moreover, we define $U(c) = 1 - \exp(-\eta \cdot c)$, where $c = \Pi(K, M, X; \tau)$. This is the exponential utility function, which is often used when risk (such as tariff uncertainty) is present. Moreover, exponential utility avoids the domain error that may be present under log utility.

The Bellman equation is then

$$V(K, \tau) = \max_{M \geq 0, X \geq 0} \left\{ U(K^\alpha [(1 - \psi)M]^\gamma - (1 + \tau)pM - X) + \beta \mathbb{E}[V(K', \tau') \mid \tau] \right\},$$

subject to

$$K' = (1 - \delta) K + X + \psi M.$$

Here

$$\mathbb{E}[V(K', \tau') \mid \tau] = P_{\tau_L} \cdot V(K', \tau_L) + P_{\tau_H} \cdot V(K', \tau_H).$$

The model is dynamic because the firm's decisions today affect not only current profits but also future capital accumulation, and thus future profits. In particular, the firm makes decisions about investment X_t and imports M_t with awareness of how these choices will shape the capital stock K' in the next period.

Moreover, the core intertemporal trade-off emerges in deciding between current profits — where higher M_t and X_t will reduce current profit — or future profits, as current investment and imports will increase future capital, and thus enhance future profits.

3 Computational Approach

3.1 Numerical Method Overview

We solve the firm's dynamic optimization problem numerically using the Bellman iteration algorithm combined with Brent's method for the continuous optimization of two control variables: imports (M) and investment (X). The solution method consists of iterating on the Bellman equation until convergence to obtain the value function and optimal policy functions.

A typical brute-force grid search over possible values for X and M requires evaluating the objective function at many discrete points, which can be computationally costly and imprecise if the grid isn't fine enough. We use Brent's method to accurately and efficiently find the minimum of the objective function. This allows us to reduce computational complexity by performing fewer function evaluations, find optimal points on a continuous range rather than a discrete grid, and ensure convergence to a precise optimum without relying on finely spaced grids.

3.2 Initial Parameters

Parameter	Description	Value
α	Capital elasticity of production	0.4
γ	Import elasticity of production	0.3
η	Risk aversion constant	0.01
δ	Depreciation rate of capital	0.1
p_M	World price of imports	1.0
τ	Tariff rate	(0, 0.4)
ψ	Share of imported goods that contribute to future capital	0.08
β	Discount factor of future profits	0.9

Table 1: Suggested Baseline Parameters

3.3 State Space and Grids

The state space is discretized using two state variables:

- **Capital** (K): We create an evenly spaced grid of capital values, defined as

$$K \in [K_{\min}, K_{\max}], \quad \text{with } n_K \text{ grid points.}$$

- **Tariff** (τ): Discretized into two states, low and high, denoted by $\tau \in \{\tau_L, \tau_H\}$.

We denote the discretized value function at iteration n by:

$$V^{(n)}(K_i, \tau_j), \quad \text{where } i = 1, \dots, n_K \text{ and } j \in \{L, H\}.$$

3.4 Value Function Approximation

To approximate the value function for continuous values of K' , we employ piece-wise linear interpolation in capital. Specifically, at each iteration, we construct linear spline interpolations:

$$sV_j(K) = \text{LinearInterpolation}(\{K_i\}, \{V^{(n)}(K_i, \tau_j)\}, \text{Line}()), \quad j \in \{L, H\}.$$

This allows us to accurately evaluate the continuation value at arbitrary points K' that are not necessarily on the grid. Moreover, it smoothes the approximation of the value function, improving numerical accuracy, and prevents needing excessively dense grids, reducing computation time.

3.5 Bellman Iteration Algorithm

Algorithm 1 Value Function Iteration with Nested Brent Optimization

- 1: Initialize the value function: $V^{(0)}(K_i, \tau_j) \leftarrow 0$.
- 2: Set tolerance $\varepsilon = 1e - 6$ and maximum iterations $N_{\max}$.
- 3: **while** iteration $n < N_{\max}$ and $\Delta > \varepsilon$ **do**
- 4: Build interpolated value functions $sV_L(K)$ and $sV_H(K)$ from previous iteration.
- 5: Initialize the iteration's convergence measure: $\Delta \leftarrow 0$.
- 6: **for** each grid point (K_i, τ_j) **do**
- 7: Define an *outer* objective function $f_X(X)$ to determine optimal investment:

$$f_X(X) = \min_{M \geq 0} \{ -[U(\Pi(K_i, M, X, \tau_j)) + \beta \mathbb{E}[sV_{\tau'}(K')]] \},$$

subject to $K' = (1 - \delta)K_i + X + \psi M$. The expectation $\mathbb{E}[sV_{\tau'}(K')]$ is computed using the Markov transition probabilities $P_{\tau, \tau'}$.

- 8: Solve the outer optimization problem via Brent's method.
- 9: After finding X_i^* , define the *inner* objective function $f_M(M)$ for optimal imports:

$$f_M(M) = -[U(\Pi(K_i, M, X_i^*, \tau_j)) + \beta \mathbb{E}[sV_{\tau'}(K')]].$$

- 10: Solve the inner optimization for imports via Brent's method.
- 11: Update the value function at (K_i, τ_j) with X_i^* and M_i^* .
- 12: Calculate the error:

$$\Delta \leftarrow |V^{(n)}(K_i, \tau_j) - V^{(n-1)}(K_i, \tau_j)|$$

- 13: **end for**
 - 14: If $\Delta < \varepsilon$, stop and report convergence. Otherwise, continue to iteration $n + 1$.
 - 15: **end while**
-

3.6 Convergence and Output

Upon convergence of the Bellman iteration (i.e., when the maximum change in the value function between iterations falls below the predefined tolerance ε), we obtain the optimal policy functions for imports $M^*(K, \tau)$, investment $X^*(K, \tau)$, and next period's capital K' , along with the corresponding value function $V(K, \tau)$.

4 Results and Discussion

4.1 Numerical Solution for the Steady States

To find the steady state of capital stock, we first fix the tariff rate. We then interpolate the discrete policy grids for X and M — to convert the discrete tables into smooth functions $s_M(K)$ and $s_X(K)$.

First define

$$f_K(K) = (1 - \delta)K + s_X(K) + \psi s_M(K) - K,$$

and solve

$$f_K(K_{ss}) = 0 \implies K_{ss} = (1 - \delta)K_{ss} + X(K_{ss}) + \psi M(K_{ss}).$$

Then

$$M_{ss} = s_M(K_{ss}), \quad X_{ss} = s_X(K_{ss}).$$

```
Policy steady-state at tau=0.00 → K* = 3.1176,  M* = 0.4385,  X* = 0.2767
Policy steady-state at tau=0.40 → K* = 2.1118,  M* = 0.1982,  X* = 0.1953
```

We observe that under low tariffs, the steady states for all K , M , and X are higher. This is expected, as lower tariffs reduce the price of imported goods, which raises the profit for any amount of M . Hence, the firm will choose a higher M and X to maximize K , and thus utility.

Under low tariff rates and low starting capital:

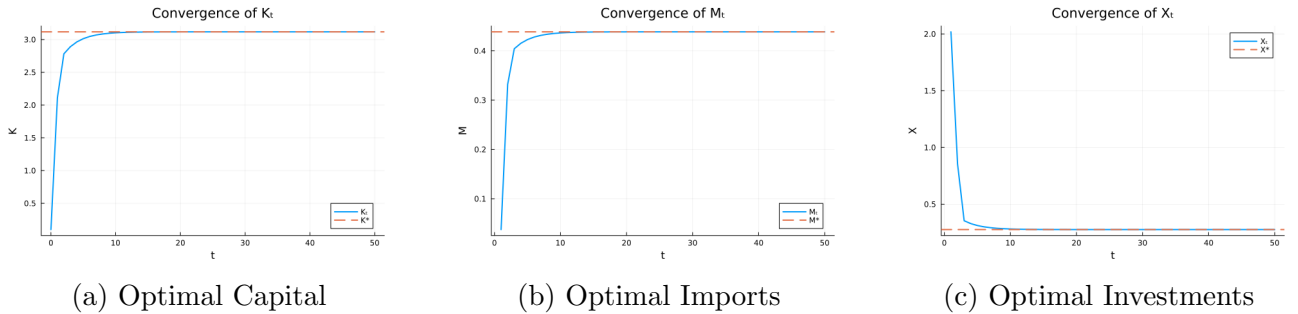
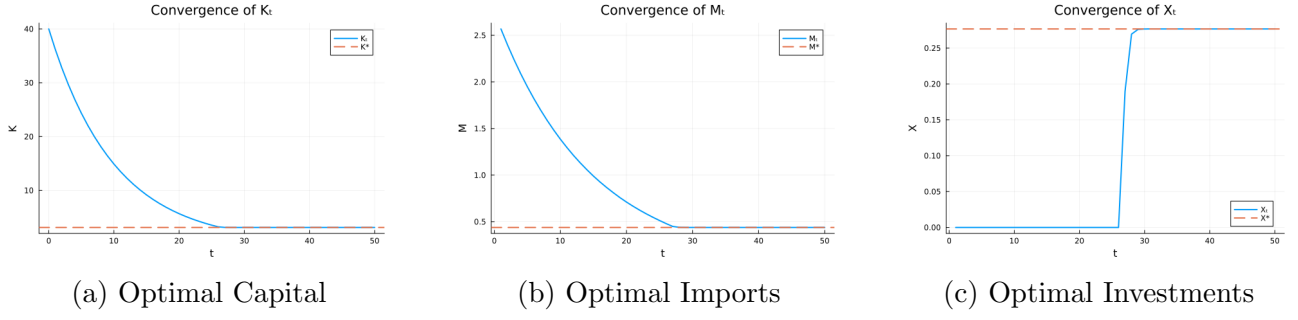
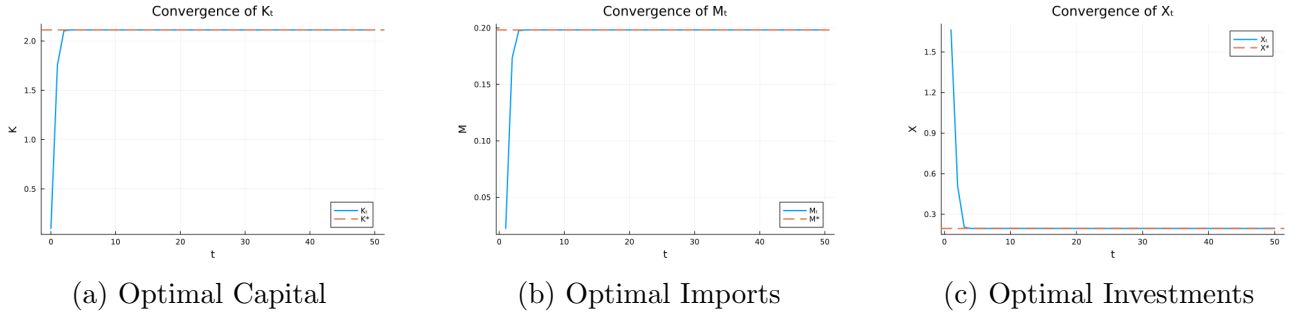


Figure 1: Variable paths under $\tau = 0$ and $K_0 = 0$

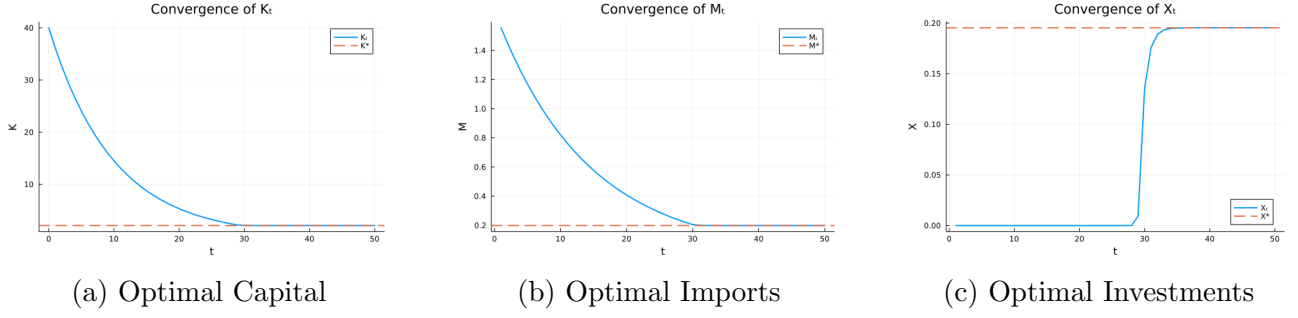
Under a zero-tariff regime, the simulated paths for capital, imports, and investment all rise quickly from a low starting point and then flatten out at their respective steady-state levels. In panel (a), the capital stock K_t jumps from its initial value and approaches $K_{ss} \approx 3.12$ by around period 10. Panel (b) shows a similar pattern for imports M_t : after an initial climb from near zero, imports settle at $M_{ss} \approx 0.44$ within the first 12 periods. In panel (c), investment X_t spikes immediately—reflecting the firm’s incentive to “catch up” when capital is scarce—and then decays rapidly to $X_{ss} \approx 0.28$. In each graph the solid blue line traces the time path while the dashed red line denotes the steady-state level.

Under low tariff rates and high starting capital:Figure 2: Variable paths under $\tau = 0$ and $K_0 = 40$

Under zero-tariff conditions and starting from a high initial capital $K_0 = 40$, the firm's policy paths exhibit a declining dynamic toward the same steady state. In panel (a), the capital stock K_t falls monotonically from its initial value toward the steady-state $K_{ss} \approx 3.12$ over roughly 30–40 periods (dashed red line). Panel (b) shows imports M_t likewise declining from an initial high level (near 2.5) down to $M_{ss} \approx 0.44$. In panel (c), investment X_t remains essentially zero while $K_t > K_{ss}$ —reflecting no incentive to invest when capital is abundant—until K_t approaches its steady state, at which point there is a one-time spike to $X_{ss} \approx 0.28$, after which investment again falls back to zero. These trajectories illustrate that, with ample initial capital, the firm allows depreciation to erode its stock and postpones both imports and new investment.

Under high tariff rates and low starting capital:Figure 3: Variable paths under $\tau = 0.4$ and $K_0 = 0$

Under a high-tariff regime and a low starting capital $K_0 = 0$, the simulated paths still converge rapidly but to lower equilibrium levels. In panel (a), the capital stock K_t rises from zero and approaches its steady state $K_{ss} \approx 2.11$ by around period 10. Panel (b) shows imports M_t increasing from almost zero to $M_{ss} \approx 0.20$ over the same horizon. In panel (c), investment X_t initially spikes—as the firm builds its capital stock—but then decays quickly and settles at $X_{ss} \approx 0.195$. Hence, it seems that costly inputs reduce the steady state outcome but do not meaningfully alter the speed of convergence under low starting capital.

Under high tariff rates and high starting capital:Figure 4: Variable paths under $\tau = 0.4$ and $K_0 = 40$

Under a high-tariff regime and a high initial capital $K_0 = 40$, the firm's decision rules exhibit a pronounced “run-down” followed by a delayed adjustment back toward the lower steady state — a trend similar to that exhibited in Figure 2. In panel (a), the capital stock K_t declines smoothly from its initial value to the steady-state level $K_{ss} \approx 2.11$ over the first 30–40 periods. Panel (b) shows imports M_t likewise falling from an initial high level (near 1.5) down to $M_{ss} \approx 0.20$. In panel (c), investment X_t remains essentially zero while $K_t > K_{ss}$, then exhibits a one-time spike up to $X_{ss} \approx 0.195$ as capital crosses its equilibrium level, after which investment returns to zero. These trajectories illustrate that with both costly inputs and abundant initial capital, the firm allows depreciation to erode its stock and postpones new imports and investment until it reaches the new high-tariff steady state, similarly to the low-tariff / high-capital regime.

4.2 Average Optimal Policies

Tariff Rate	Average Optimal Imports	Average Optimal Investment
$\tau = 0.0$	1.5183	0.0940
$\tau = 0.4$	0.9157	0.0520

Table 2: Numerical Solutions to $\bar{M}$ and $\bar{X}$

The Bellman iteration converged after 79 iterations. Clearly, the results show that under low tariffs, both imports and investment tend to be higher. This makes sense, as low tariff rates make imports cheaper, so the marginal return of imports is higher (as they increase future capital), so the firm has more incentives to import more. Moreover, investments are higher, as the marginal return of capital is higher (aided by cheaper imports, and hence higher profits), which incentivizes the firm to invest more to increase future capital.

4.3 Value Function

Figure 5 shows that both value functions $V(K, \tau)$ are strictly increasing and concave in capital K , reflecting diminishing marginal returns: as the firm accumulates more capital, its discounted lifetime profit rises but at a diminishing rate. At $K = 0$, the two value functions coincide (about 0 units), since with zero capital the firm cannot produce or import inputs. As K grows, the blue curve lies above the red one by an expanding margin, reflecting the fact that a 40% tariff raises input costs and therefore reduces the present value of all future profits. When capital

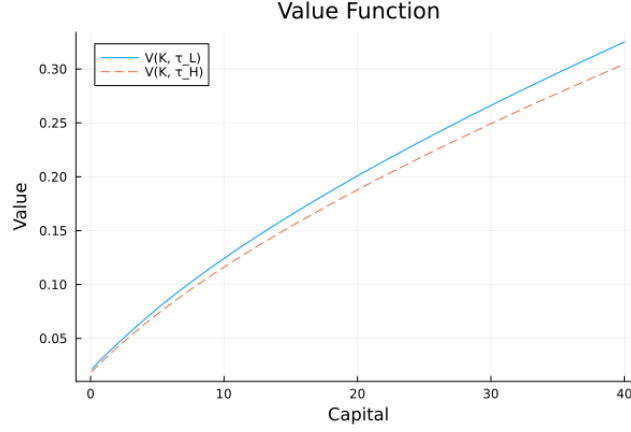


Figure 5: Value Function

reaches 40, the low-tariff value is roughly 0.30 while the high-tariff value is about 0.32, a gap of approximately 0.02 value units. This is expected, as higher tariffs increase the per-unit price of intermediate imports, decreasing the profit of the firm in a given period.

4.4 Optimal Import Policy

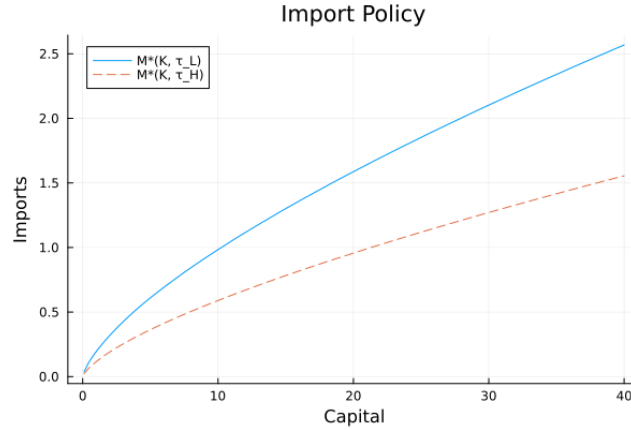


Figure 6: Import Policy

Under both tariff rates, imports increase in K but at a decreasing rate—larger capital stocks make importing more profitable, yet the marginal payoff to each additional unit of import falls. Crucially, the high-tariff curve lies uniformly below the low-tariff curve and the gap between optimal imports widens as capital stock increases. At $K = 10$, optimal imports are approximately 40% lower under high tariffs as compared to low tariffs ($M \approx 1.0$ under $\tau = 0$ and $M \approx 0.6$ under $\tau = 0.4$). However, at $K = 40$, optimal imports are around 68% lower under high tariffs than low tariffs (at $K = 40$ — the firm would import about 2.6 units when $\tau = 0$ but only about 1.5 units when $\tau = 0.4$).

This makes sense. At low levels of capital, the base productivity of imports is likely low, so tariffs have a limited absolute impact. However, at high levels of capital, imports are likely more productive as the firm has more capital to use them efficiently, but the higher tariffs reduce this productivity gain and reduce the incentive to import. With higher tariffs, the firm must weigh

higher import costs against their marginal productivity (how much they contribute towards future capital gain). This cost grows disproportionately at high K , leading to a widening gap.

4.5 Optimal Investment Policy

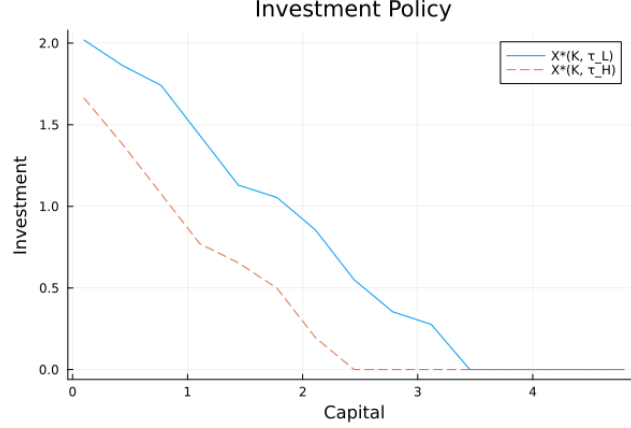


Figure 7: Investment Policy

For the optimal investment policy X^* , we focus on the first 15 values of K , since after that the optimal investment under both high and low tariff rates is approximately the same (at 0). Nonetheless, we observe that investment is large when K is low and declines until it hits zero. Under $\tau = 0$, investment falls to zero at about $K \approx 3.5$, whereas under $\tau = 0.4$ it stops already at $K \approx 2.3$. Beyond those points, the firm makes no positive investment and simply lets depreciation erode its capital stock back toward the build-up level. Average investment likewise falls by roughly half under tariffs ($\bar{X}(0) \approx 0.094$ vs. $\bar{X}(0.4) \approx 0.052$ from Table 2).

Hence, it seems that investment starts out higher under the low-tariff regime because the shadow value of one more unit of capital — i.e. the slope of the value function — is steeper when future profits are less attenuated by tariffs. Thus the firm has stronger incentives to devote resources toward investment when $\tau = 0$, leading to a larger positive value in X at low K .

4.6 Optimal Capital Policy

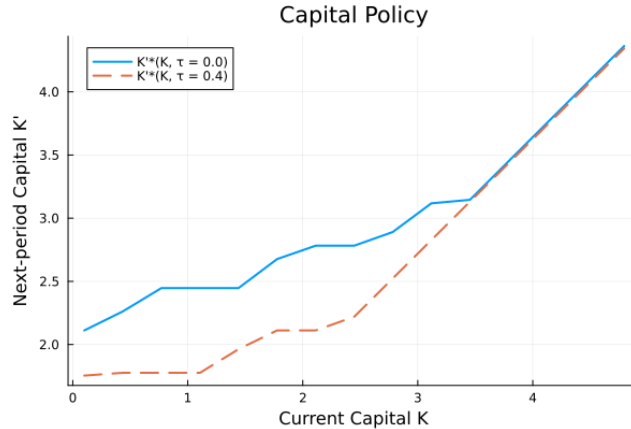


Figure 8: Capital Policy

For the optimal capital policy plot in Figure 8, we again only observe the first 15 values of K . At very low current capital levels, we note that under low tariffs $\tau = 0$ importing is cheaper, and so the firm sees that higher imports will boost current output via $K^\alpha M^\gamma$ and future capital via ψM . Therefore, when imports are cheap, it is more profitable for the firm to target a larger capital "build-up" in K' (at around ≈ 3.45) because future returns to capital are larger. Under high tariffs $\tau = 0.4$, however, importing becomes more expensive, and so the firm will import less, reducing current output and future capital. The incentive to accumulate a larger K' is now weaker, since capital build-up is harder and profits are diminished through increased per-unit cost of imported goods. Hence, the firm will choose a lower "build-up" target (at around ≈ 2.45), as it is not worth it to increase imports for small and costly increases in future capital.

At around $K \geq 3.5$, however, we see that both optimal capital policies essentially converge. This happens because of the diminishing returns to capital (depicted in the concavity of the value function). Once the firm has accumulated "enough" capital, the marginal value of capital is not profitable enough and the tariff-induced difference in future returns becomes small, hence the similar behavior under low and high tariff rates — the firm simply replaces depreciated capital stock. This is also observed in the capital law of motion: since optimal M is small, and only ψM influence K' , then when optimal $X = 0$, the next-period's capital are not significantly influenced by tariffs.

4.7 Optimal Path Simulations

We set initial capital at $K = 2.0$ and define

$$P = \begin{pmatrix} 0.85 & 0.15 \\ 0.10 & 0.90 \end{pmatrix}.$$

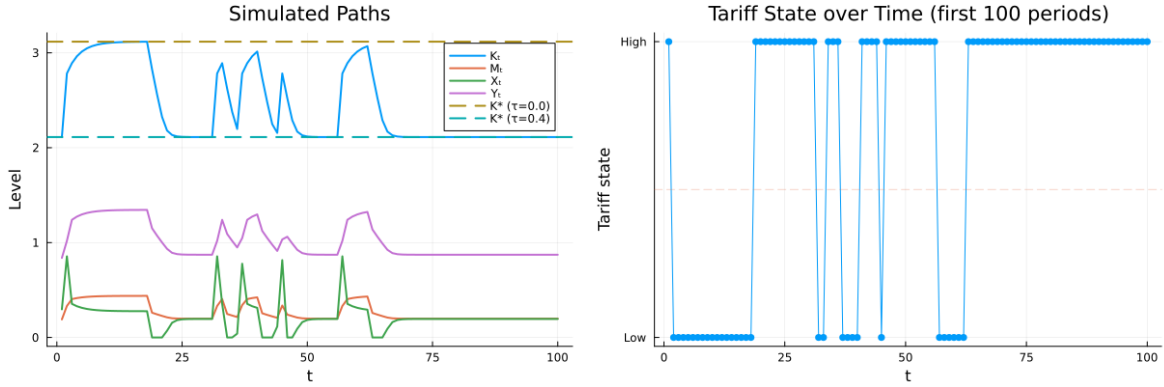


Figure 9: Simulated Economic Dynamics and State Transitions (first 100 periods)

Initially, the firm builds up capital quickly from low levels — matching the optimal capital policy (where small K leads to big jumps). After reaching the target steady-state level, K_t becomes almost flat — just replacing depreciation — under both regimes. Fluctuations come from tariff changes (high tariffs lower investment/imports, causing capital to slowly decay or grow less). In general, we see that high tariffs are more persistent, which agrees with the Markov process matrix.

Moreover, we see that M_t is sensitive to tariff rates. When $\tau = 0.4$, M_t drops sharply, and recovers when $\tau = 0$ more slowly. This makes sense, as we previously saw that high tariffs make imports less attractive. X_t is also highly sensitive to tariff rates, as it stabilizes at around $X \approx 0.2$ under $\tau = 0.4$, but increases sharply when $\tau = 0$. However, even when $\tau = 0$, X quickly drops after the initial jump. This can be interpreted as the firm investing heavily to build up to the target steady-state K , and then simply investing enough to offset depreciation — i.e. not investing enough to actually increase future capital. Also, we see that when transitioning from $\tau = 0$ to $\tau = 0.4$, X drops to 0, as incentives to invest collapse. Finally, because production Y_t depends on both K_t and M_t , and K_t moves more smoothly while M_t drops more sharply under high tariffs, then we see that Y_t follows the general trend for K_t and M_t .

5 Conclusion and Outlook

In this project, we developed and solved a dynamic firm optimization model to study the effects of tariff uncertainty on production decisions. Using a Cobb–Douglas production function, dynamic capital accumulation with learning-by-importing, and a stochastic tariff environment, we demonstrated how firms can optimally adjust imports and investments when faced with changes in trade policy. The model, solved via Bellman iteration with nested Brent optimization, revealed that lower tariffs encourage both higher imports and greater initial investment, while higher tariffs reduce incentives to expand and lead firms to prioritize maintaining existing capital rather than accumulating new capital. Simulation of firm behavior over time confirmed the predicted dynamics, with capital stabilizing at higher levels under low tariffs and output closely tracking shifts in the tariff regime.

Importantly, the model shows how external policy shocks can propagate into firms’ long-run growth paths by influencing their intertemporal decisions. In periods of low tariffs, firms invest more heavily to build future capital stock. In periods of high tariffs, however, firms reduce imports and investments, ultimately achieving lower value in the long run. The results illustrate that the effects of tariffs are not limited to static distortions of input prices but can cause lasting dynamic inefficiencies by reducing future productive capacity (capital) and output. Through this computational approach, we gain insight into how protectionist policies may unintentionally hurt domestic productivity and growth by discouraging firms from upgrading their capital stock.

Moreover, the findings of this project have direct relevance to real-world trade tensions and industrial policy debates. In recent years, tariff wars — notably between the United States and China — have reignited concerns over how tariffs affect domestic firms. Empirical studies show that tariffs imposed during the 2018–2020 period were largely passed onto U.S. importers and consumers, raising input costs and discouraging investment in sectors reliant on global supply chains^[2]. The model developed here captures this dynamic: firms facing unpredictable tariff shocks respond by reducing their reliance on imported intermediate goods and by scaling back investment, resulting in lower long-run productivity. As the global economy grapples with persistent supply chain disruptions and a rethinking of globalization, understanding these intertemporal effects is crucial for designing trade policies that balance short-term strategic objectives with long-term economic vitality.

References

- [1] Wikipedia. Chicken tax. https://en.wikipedia.org/wiki/Chicken_tax, 2025.
- [2] Erica York. Trump tariffs: Tracking the economic impact of the trump trade war. <https://www.wita.org/atp-research/tracking-impact-trump-tariffs/>, March 2025.

6 Appendix

6.1 Bellman Iteration Code

```

# Parameters
alpha, gamma = 0.5, 0.3      # Cobb{Douglas exponents
beta, delta = 0.9, 0.1      # discount factor and depreciation
p_M           = 1.0          # world price of imports
psi           = 0.08         # share of imports that augment capital
eta = 0.01                 # risk aversion constant (mild)

# Grids
K_min, K_max, nK = 0.1, 40.0, 120
K_grid = collect(range(K_min, K_max, length = nK))

_L, _H = 0.0, 0.4
tau_grid = [_L, _H]
P_tau    = [0.85 0.15;
            0.10 0.90]
n_tau    = length(tau_grid)

# Bracketing intervals for Brent searches
X_lo, X_hi = 0.0, 10.0
M_lo, M_hi = 0.0, 10.0

# Helper functions
makespline(a, b)              = LinearInterpolation(a, b, extrapolation_bc = Line())
production(K, M)              = K^alpha * ((1 - psi)*M)^gamma
flow_profit(K, M, X, ) = production(K, M) - (1 + )*p_M*M - X
utility(c)                   = 1-exp(-eta * c)                # exponential utility
k_prime(K, M, X)             = (1 - delta)*K + X + psi*M

# Bellman Iteration
V          = zeros(nK, n_tau)
Vnext      = similar(V)
policy_M   = zeros(nK, n_tau)
policy_X   = zeros(nK, n_tau)

tol, max_iter = 1e-6, 1000
iter, bellman_err = 0, Inf

while iter < max_iter && bellman_err > tol
    iter += 1

    # // reset error for this pass //
    bellman_err = 0.0

    # build interpolants of value function

```

```

sV = [makespline(K_grid, V[:, j]) for j in 1:n_tau]

for (iK, K) in enumerate(K_grid), (jtau, tau) in enumerate(tau_grid)

    # outer maximization over X
    fX = X -> begin
        # inner maximization over M
        fM = M -> begin
            profit = flow_profit(K, M, X, tau)
            Kp = k_prime(K, M, X)
            EV = P_tau[jtau,1]*sV[1](Kp) + P_tau[jtau,2]*sV[2](Kp)
            return -(utility(profit) + beta*EV)
        end

        resM = optimize(fM, M_lo, M_hi, Brent(); rel_tol=1e-6)
        return Optim.minimum(resM)
    end

    # find optimal X
    resX = optimize(fX, X_lo, X_hi, Brent(); rel_tol=1e-6)
    val_neg, X_star = Optim.minimum(resX), Optim.minimizer(resX)

    # recover optimal M at X_star
    fM_star = M -> begin
        profit = flow_profit(K, M, X_star, tau)
        Kp = k_prime(K, M, X_star)
        EV = P_tau[jtau,1]*sV[1](Kp) + P_tau[jtau,2]*sV[2](Kp)
        return -(utility(profit) + beta*EV)
    end

    resM = optimize(fM_star, M_lo, M_hi, Brent(); rel_tol=1e-6)
    M_star = Optim.minimizer(resM)

    # store results
    best_val = -val_neg
    Vnext[iK,jtau] = best_val
    policy_M[iK,jtau] = M_star
    policy_X[iK,jtau] = X_star

    # get error
    bellman_err = max(bellman_err, abs(best_val - V[iK,jtau]))
end

# update
V .= Vnext
if iter % 20 == 0
    @printf("iter %4d : err = %.3e\n", iter, bellman_err)
end
end
println("Converged in $iter iterations. Final error: $bellman_err")

```

6.2 Plotting Code

```

n_points = 15
policy_Kp = [ k_prime(K_grid[i], policy_M[i,s], policy_X[i,s])
              for i in 1:nK, s in 1:n_tau ]

pM = plot(
    K_grid, policy_M[:,1],
    label="M*(K, _L)",
    xlabel="Capital",
    ylabel="Imports",
    title="Import Policy")
plot!(pM, K_grid,
      policy_M[:,2],
      label="M*(K, _H)", ls=:dash)
savefig(pM, "final_paper/plots/imports.png")

pX = plot(
    K_grid[1:n_points], policy_X[1:n_points,1],
    label="X*(K, _L)",
    xlabel="Capital",
    ylabel="Investment",
    title="Investment Policy")
plot!(pX, K_grid[1:n_points],
      policy_X[1:n_points,2],
      label="X*(K, _H)", ls=:dash)
savefig(pX, "final_paper/plots/investment.png")

pKp = plot(
    K_grid[1:n_points], policy_Kp[1:n_points, 1],
    label = "K'*(K, = $(tau_grid[1]))",
    xlabel = "Current Capital K",
    ylabel = "Next-period Capital K'",
    title = "Capital Policy",
    lw = 2
)
plot!(
    pKp,
    K_grid[1:n_points], policy_Kp[1:n_points, 2],
    label = "K'*(K, = $(tau_grid[2]))",
    ls = :dash,
    lw = 2)
savefig(pKp, "final_paper/plots/capital_policy.png")

pV = plot(K_grid, V[:,1], label="V(K, _L)",
          xlabel="Capital", ylabel="Value", title="Value Function")
plot!(pV, K_grid, V[:,2], label="V(K, _H)", ls=:dash)
savefig(pV, "final_paper/plots/value_function.png")

```


6.3 Average Optimal Policies

```

avg_M = [mean(policy_M[:, s]) for s in 1:n_tau]
avg_X = [mean(policy_X[:, s]) for s in 1:n_tau]

println("\nAverage optimal imports and investment:")
for s in 1:n_tau
    @printf("    = %.3f : M = %.4f X = %.4f\n",
            tau_grid[s], avg_M[s], avg_X[s])
end

```

```

Average optimal imports and investment:
= 0.000 : M = 1.5183 X = 0.0940
= 0.400 : M = 0.9157 X = 0.0520

```

6.4 Numerical Solution to Steady States Code

```

tau_grid = [0.0, 0.4]
= 0.4
jtau = findfirst(==( ), tau_grid)

sM = LinearInterpolation(K_grid, policy_M[:, jtau], extrapolation_bc=Line())
sX = LinearInterpolation(K_grid, policy_X[:, jtau], extrapolation_bc=Line())

# root finding
fK(K) = (1 - delta)*K + sX(K) + psi*sM(K) - K
K_star = find_zero(fK, (minimum(K_grid), maximum(K_grid)))
M_star = sM(K_star)
X_star = sX(K_star)

@printf("Policy steady-state at =%.2f → K* = %.4f, M* = %.4f, X* = %.4f\n",
        , K_star, M_star, X_star)

function simulate(K0; T=50)
    K_path = zeros(T+1); M_path = zeros(T); X_path = zeros(T)
    K_path[1] = K0
    for t in 1:T
        K = K_path[t]
        M = sM(K)
        X = sX(K)
        M_path[t] = M; X_path[t] = X
        K_path[t+1] = (1 - delta)*K + X + psi*M
    end
    return K_path, M_path, X_path
end

```

```

# change initial capital
K0 = maximum(K_grid)
T = 50
K_path, M_path, X_path = simulate(K0, T=T)

# plot
ts = 0:T
p1 = plot(ts, K_path, lw=2, label="K",
          xlabel="t", ylabel="K", title="Convergence of K")
hline!([K_star], ls=:dash, lw=2, label="K*")
savefig(p1, "final_paper/plots/converge_K_exact.png")

p2 = plot(1:T, M_path, lw=2, label="M",
          xlabel="t", ylabel="M", title="Convergence of M")
hline!([M_star], ls=:dash, lw=2, label="M*")
savefig(p2, "final_paper/plots/converge_M_exact.png")

p3 = plot(1:T, X_path, lw=2, label="X",
          xlabel="t", ylabel="X", title="Convergence of X")
hline!([X_star], ls=:dash, lw=2, label="X*")
savefig(p3, "final_paper/plots/converge_X_exact.png")

Policy steady-state at =0.40 → K* = 2.1118, M* = 0.1982, X* = 0.1953

```

6.5 Optimal Path Simulations Code

```

# redefine spline to avoid errors
makespline(a::AbstractVector, b::AbstractVector) =
    LinearInterpolation(a, b, extrapolation_bc = Line())

# arrays of callable interpolants
n_tau = length(tau_grid)
m_interp = [ makespline(K_grid, policy_M[:,s]) for s in 1:n_tau ]
x_interp = [ makespline(K_grid, policy_X[:,s]) for s in 1:n_tau ]

K_star = zeros(n_tau)
for s in 1:n_tau
    fK(K) = (1 - delta)*K + x_interp[s](K) + psi*m_interp[s](K) - K
    K_star[s] = find_zero(fK, (minimum(K_grid), maximum(K_grid)))
end

T, T_plot = 1000, 100
rng = MersenneTwister(42) # random number generator

k_path = zeros(T+1)

```

```

m_path = zeros(T)
x_path = zeros(T)
y_path = zeros(T)
state = zeros(Int, T+1)

# initial values
k_path[1] = 2.0
state[1] = rand(rng) < 0.5 ? 1 : 2

for t in 1:T
    K_cur = k_path[t]
    s_cur = state[t]

    # call to find optimal values
    M_star = m_interp[s_cur](K_cur)
    X_star = x_interp[s_cur](K_cur)

    y_path[t] = K_cur^alpha * ((1-psi)M_star)^gamma
    m_path[t] = M_star
    x_path[t] = X_star

    # law of motion
    k_path[t+1] = (1 - delta)*K_cur + X_star + psi*M_star

    # tariff transition
    u = rand(rng)
    state[t+1] = (u < P_tau[s_cur,1]) ? 1 : 2
end

# plot
ts = 1:T_plot
p = plot(ts, k_path[1 .+ ts], label="K", lw=2,
        xlabel="t", ylabel="Level", title="Simulated Paths")
plot!(p, ts, m_path[ts], label="M", lw=2)
plot!(p, ts, x_path[ts], label="X", lw=2)
plot!(p, ts, y_path[ts], label="Y", lw=2)
for s in 1:n_tau
    hline!(p, [K_star[s]],
        label="K* (=$(tau_grid[s]))", ls=:dash, lw=2)
end
plot!(legend=:topright, grid=true)
savefig(p, "final_paper/plots/simulated_paths.png")

plt_state = plot(ts, state[ts],
    xlabel="t",
    ylabel="Tariff state",
    yticks=( [1,2], ["Low","High"] ),
    markershape=:circle,
    markerstrokewidth=0,
    legend=false,

```

```
    title="Tariff State over Time (first $T_plot periods)" )  
hline!([1.5], ls=:dash, alpha=0.3)  # optional midpoint guide  
savefig(plt_state, "final_paper/plots/state_transitions.png")
```